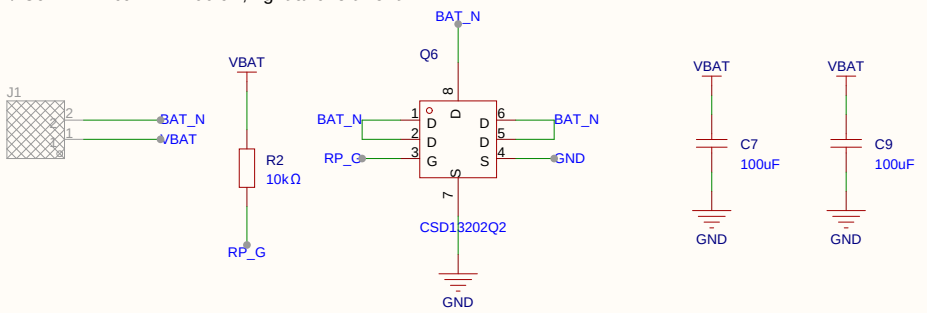
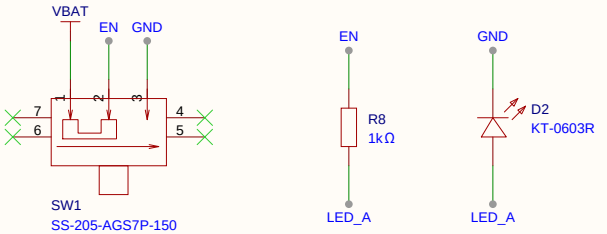
BATTERY INPUT AND REVERSE PROTECTION

Q6/R2: low-side FET in the battery return; reversed battery -> no power, LED off
C7 / C9: VBAT bulk 2 x 100 uF, right and left front



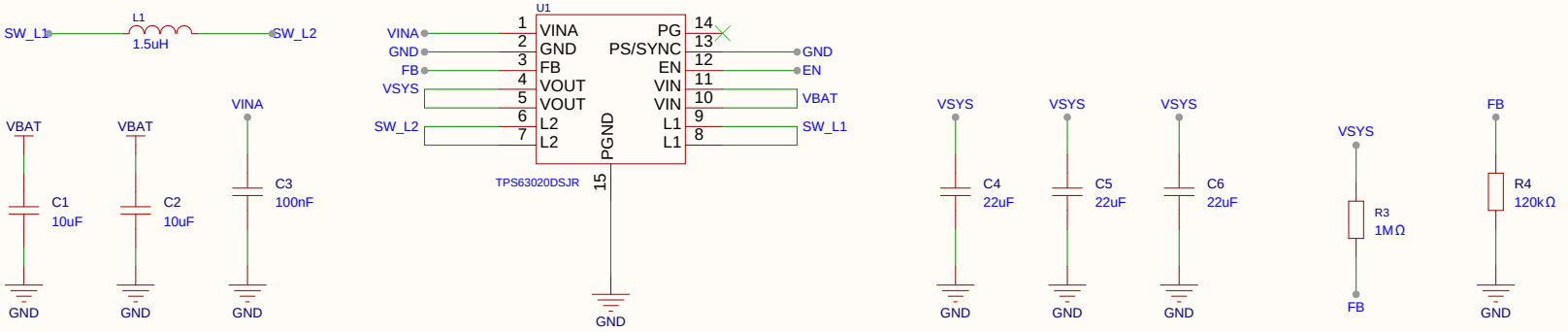
POWER SWITCH

SW1: slide toward pin 1 (VBAT) = ON; carries only the EN current
D2: switch LED on EN: on when SW1 is on and the battery is connected; USB alone does not light it



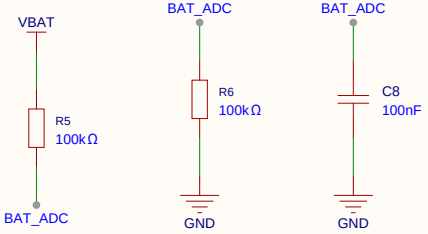
4.67 V BUCK-BOOST (TPS63020)

VSYS = 0.5 x (1 + R3/R4) = 4.67 V; PS/SYNC low = power-save; PG unused
VINA: only C3 100 nF to GND (TI rev G, internal VIN filter); C1/C2 at VIN, C4-C6 at VOUT



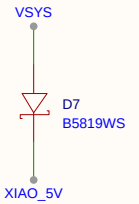
BATTERY VOLTAGE SENSE

BAT_ADC -> XIAO D0 (GPIO1, ADC1_CH0): VBAT / 2



5 V TO THE XIAO

D7: VSYS -> XIAO 5V pin (VBUS); USB 5V cannot flow back into VSYS



Schematic	Schematic1	Create at	2026-10-08
		Update at	2026-11-04
Board	Board1	Page	P1-Power
Drawn		sparrow-fc-v1.0	
Reviewed			
		Version	Size / Page 1 Total 3
EasyEDA		V1.0	A4 / EasyEDA.com

XIAO ESP32-S3 PLUS (J2, J3)

Soldered directly into two 1x7 rows (J2 = D0-D6, J3 = 5V/GND/3V3/D10-D7); sockets lose I2C readings
Powered through its 5V pin from VSYS (D7); its 3V3 pin powers GY-521 and GY-530



I2C SENSORS (J4, J9)

J4 GY-521 (MPU-6050, I2C 0x68) and J9 GY-530 (VL53L0X, I2C 0x29) share one I2C bus
Pull-ups are on the modules



GPIO MAP

Pin	GPIO	Signal
D0	GPIO1	BAT_ADC
D1	GPIO2	MOT3
D2	GPIO3	IMU_INT
D3	GPIO4	MOT4
D4	GPIO5	I2C_SCL
D5	GPIO6	I2C_SDA

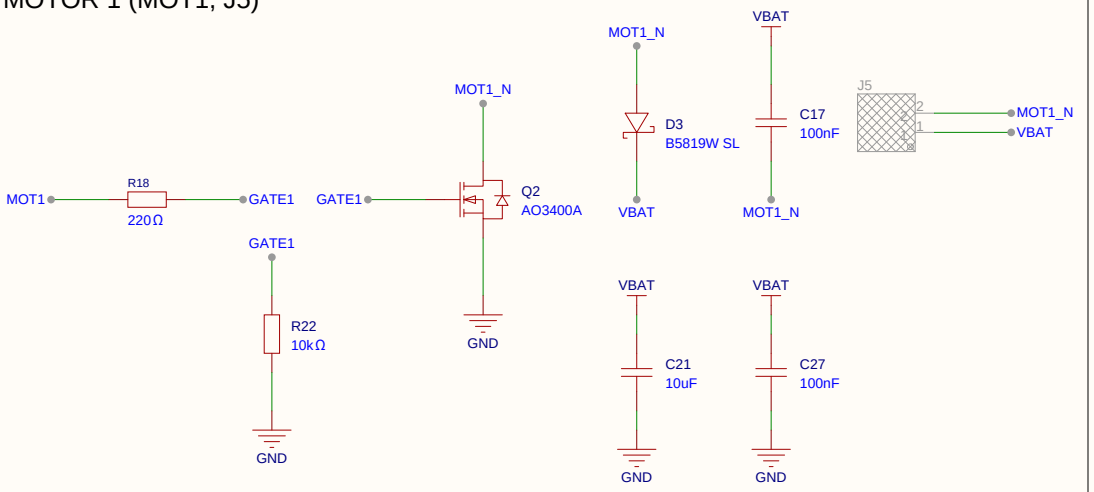
Pin	GPIO	Signal
D6	GPIO43	unused (boot log)
D7	GPIO44	TOF_XSHUT
D8	GPIO7	MOT1
D9	GPIO8	MOT2
D10	GPIO9	TOF_INT

Schematic	Schematic1		Create at	2026-10-08
			Update at	2026-11-03
Board	Board1		Page	P2-MCU-Sensors
Drawn		sparrow-fc-v1.0		
Reviewed				
			Version	Size	Page 2 Total 3
EasyEDA			V1.0	A4	EasyEDA.com

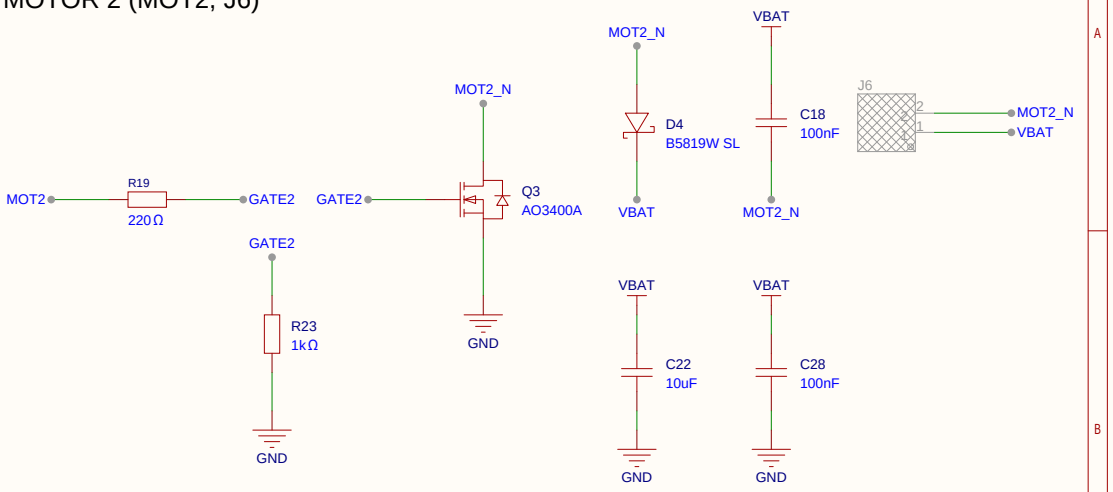
4 x LOW-SIDE MOTOR DRIVERS

AO3400A low-side switches for brushed 720 coreless motors, PWM 20-32 kHz from LEDC
Gate 220 R series + 10k pull-down; B5819W flyback (K to VBAT); 100 nF across each motor; 10 uF + 100 nF VBAT-GND per driver
R23 (motor 2) is 1k, not 10k: it keeps Q3 off against the 4.7k pull-up on D9 of the XIAO Sense camera board

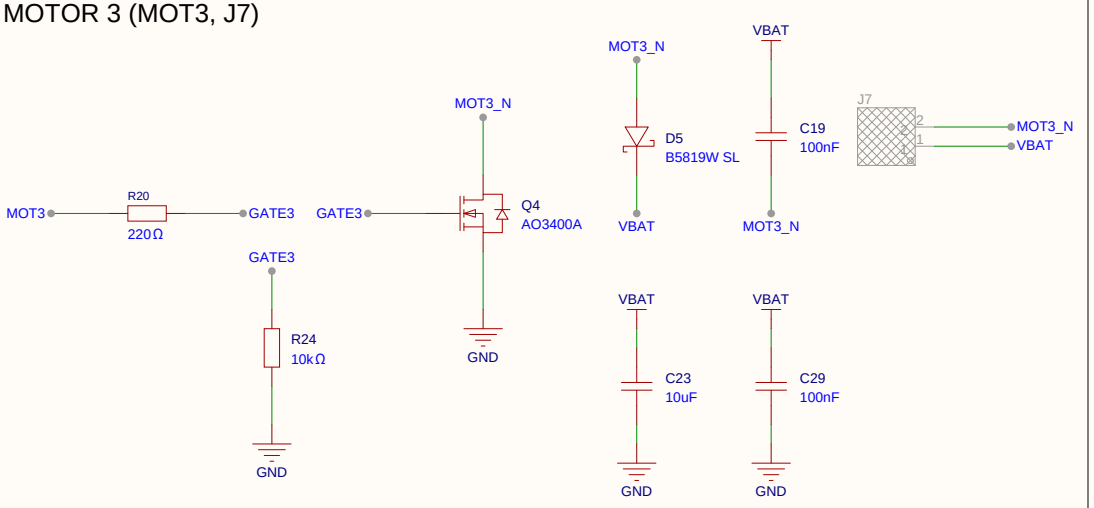
MOTOR 1 (MOT1, J5)



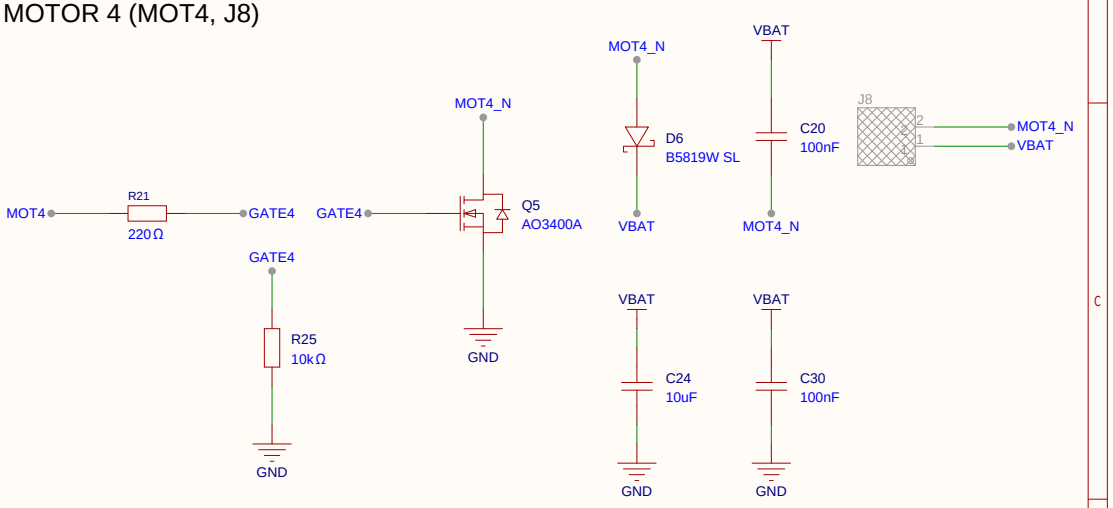
MOTOR 2 (MOT2, J6)



MOTOR 3 (MOT3, J7)



MOTOR 4 (MOT4, J8)



MOTOR PADS (J5-J8)

Pin 1 (square) = VBAT (+), pin 2 = drain
M1 rear-left CCW, M2 front-left CW
M3 front-right CCW, M4 rear-right CW

Schematic	Schematic1	Create at	2026-10-08
		Update at	2026-11-04
Board	Board1	Page	P3-Motors
Drawn		sparrow-fc-v1.0	
Reviewed			
		Version	Size
EasyEDA		V1.0	A4
		Page 3 Total 3	
		EasyEDA.com	